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SRCH

The Storage Revolution in Chile

Regulatory Design, Investment, and Wholesale Market Effects

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The puzzle

Three observations about Chile's storage deployment that are hard to reconcile with a cost or technology explanation:

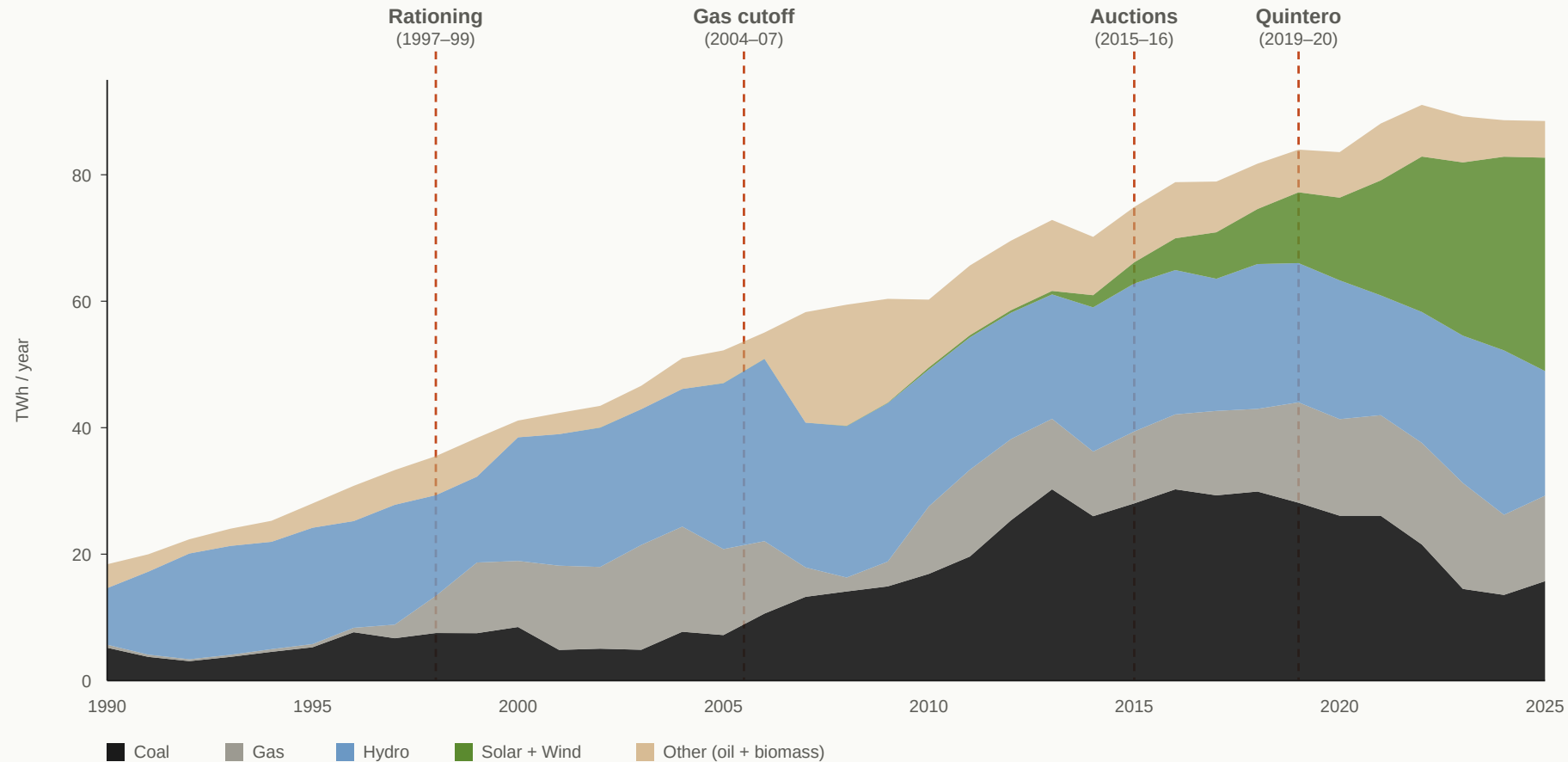
- 1 Chile crossed **2 GW of BESS in 24 months** — four years ahead of its official 2030 target.
- 2 It did so **without fiscal subsidies**, feed-in tariffs, or guaranteed revenues — the deployment was driven by market-based mechanisms.
- 3 The expansion produced **observable effects on wholesale prices**: day–night spreads compressed, evening ramps softened, and previously unbankable financial structures became viable.

"The economic viability of modular storage technologies is primarily a regulatory design problem rather than a technological or cost problem."

— Moreno, Pardow & Villamarín, *SRCH* — *The Storage Revolution in Chile*

A power system that quadrupled — and two strategic bets

Between 1990 and 2025, Chile's electricity generation rose from 18 to 89 TWh per year. The composition tracks two sequential bets, each shaped by external shocks. The first is **a thermal bet**: after the rationing of 1997–99, hydrological risk drives diversification toward natural gas and coal. The Argentine gas cutoff (2004–07) doubles down on coal as firm baseload. The second is **a green bet**: the supply auctions of 2015–16 — with hourly blocks and partial bids — pull ERNC in on price (the carrot), while the Quintero poisoning (2019–20) pushes coal out on public opinion (the stick). Solar and wind displace coal from the top.



Source: Ember Yearly Electricity Data (2026) via Our World in Data. Thermal split for 1990–1999 estimated from total minus hydro, applying historical shares (coal-dominated to 1996; Argentine gas ramp-up 1997–1999).

Five markets, five institutional regimes

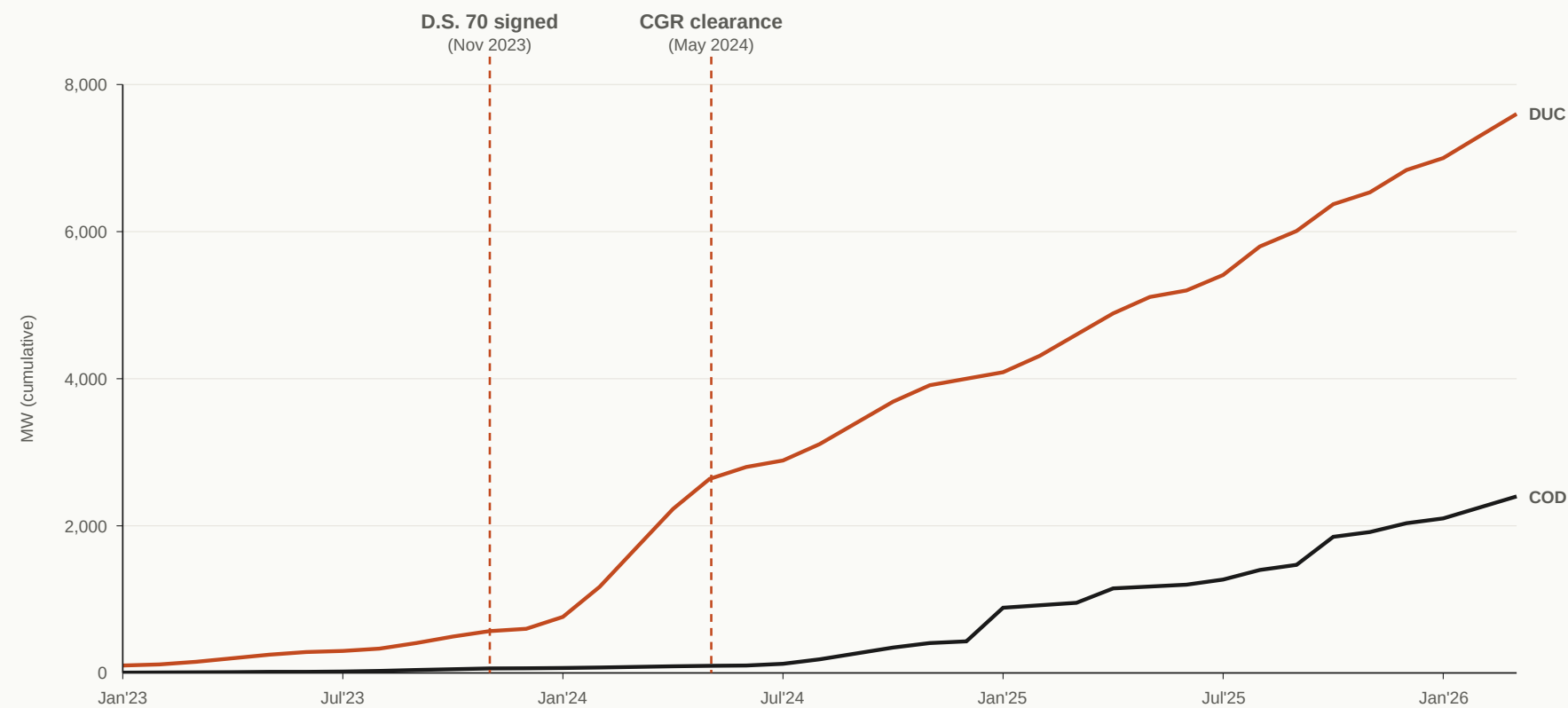
The same modular asset class operates in five distinct generation markets. **What separates the markets is not the technology — it is the regulatory regime that prices each one.** That regime determines whether the asset earns revenue, displaces costs, or remains commercially unviable.

Market	Institutional regime	Outcome
Hybrid	Wholesale + capacity payments	Taking off — co-location with renewable plants captures both energy arbitrage and capacity revenue.
Stand-alone	Wholesale + capacity payments	Taking off — merchant arbitrage now bankable in both barras; capacity remuneration anchors the case.
Transmission	Centralized planning	Stalled — the Parinas–Lo Aguirre auction was declared void in Nov 2023 and removed from the 2024 transmission plan. The problem is not asymmetric regulation between markets — it is that Tx operates as centralized planning rather than as a market. Assets enter by administrative assignment with predefined service and price ceilings, foreclosing the modular logic that makes BESS bankable elsewhere.
Distribution	Utility regulation, no passthrough	Anomalous adoption — utilities install BESS where CAPEX is lower than diesel OPEX. The Dx absorbs the full cost because it pays off internally, even without a tariff mechanism.
BTM	Retail tariffs + net billing	Open question — whether the spread between Dx purchase price and resale price suffices, absent an Australian-style rooftop incentive structure.

Five markets, four regimes — and the markets where BESS thrives are the ones structured as markets, not as central planning.

Storage took off in 24 months

DUC tracks the pipeline — projects declared under construction before the CNE. **COD** tracks the assets actually injecting. The gap between the two curves is the construction cycle, but its shape is not innocent: **DUC accelerated before the regulation was politically safe**. In a market where day-night spreads compress as storage scales, the first to reach COD captures the upside for longer — so regulatory risk and cannibalization risk pull in opposite directions, and each developer resolved that trade-off according to its own risk tolerance. **From under 100 MW in early 2023, Chile crossed 2 GW four years ahead of the official 2030 target — and the pipeline is six times larger.**



Sources: monthly anchors from CNE Reporte Energía Abierta Ciudadana, CEN Informes Mensuales SEN, Ministerio de Energía Reportes de Mercados Eléctricos, ACERA, GPM, Generadoras de Chile boletines. Monthly series interpolated between verified anchor points.

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Three features of the Chilean regulatory experience

Three features of the regulatory framework — none of them about technology or cost — explain why Chile mobilized over 2 GW of storage in 24 months while regional peers did not.

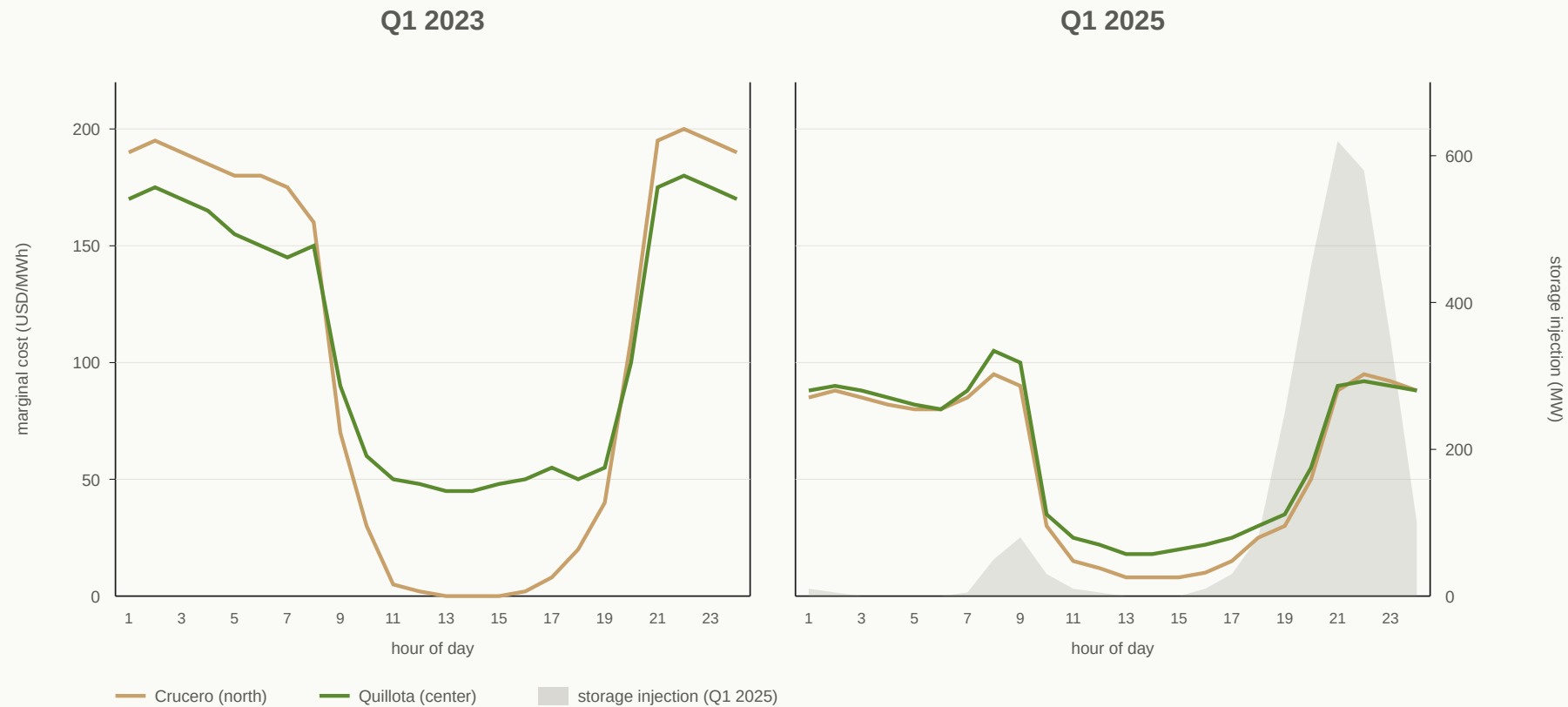
First, intertemporal consistency. The key storage reforms were initiated under one administration and implemented and scaled under another. Law 21,505 was enacted in November 2022 under one government; D.S. 70 was signed in November 2023 and cleared by Contraloría in May 2024 under the next. The coordination and operation regulations — D.S. 125 (SEN coordination) and D.S. 88 (small distributed generation) — were initiated under the previous government and continue to advance under the current one. Storage investors faced a regulatory framework that survived a change of government across multiple instruments — which mattered for the cost of capital, and ultimately for which projects reached financial close before cannibalization made later entry less attractive.

Second, regulatory flexibility. The framework adapted to the non-site-specific and reconfigurable nature of BESS in two ways. Environmental permitting allows storage to be added to existing renewable projects without restarting full approval processes, saving years in development timelines. Electrical classifications recognize storage as a distinct asset class that can participate simultaneously in energy arbitrage and capacity payments — the dual revenue stream that made the merchant case bankable from 2024 onwards.

Third, market-based incentives. Chile did not deploy direct fiscal subsidies, feed-in tariffs, guaranteed revenues, or transmission-tariff stampings to attract storage. The signals that mobilized over 2 GW in operation and 7 GW in pipeline came from the wholesale market — price spreads between solar and non-solar hours — and from capacity payments calibrated to system value. Fiscal exposure was minimal; the distributional politics of cross-subsidies was mostly avoided.

The spread collapsed — and storage moved into the gap

In **Q1 2023**, a merchant project facing daytime prices near zero couldn't run a corporate finance case, and a PPA was equally unworkable — off-takers would have to absorb extreme non-solar prices. In practice, only portfolio firms with balance-sheet capacity could underwrite Crucero. By **Q1 2025**, the night peak is roughly half and the daytime trough has lifted off zero. **Both merchant corporate finance and PPA structures become viable at the same time, in both barras.** The financially universe expands sharply — which is what unlocks the second wave of pipeline.



The Q1 2023 profile selected for portfolio firms with balance-sheet capacity. The Q1 2025 profile selects for execution speed.

Sources: hourly marginal cost profiles derived from Ministerio de Energía Reportes de Mercados Eléctricos (RME) Q1 2023 and Q1 2025, complemented with Valgesta Nueva Energía monthly bulletins. Storage injection profile based on CEN operational data 2025.

Five lessons for Latin America

What the Chilean experience suggests for other countries seeking to integrate storage as part of their decarbonization strategies — and where caution is warranted before assuming the model transfers.

- 1 Regulatory consistency across political cycles materially affects investment outcomes.**
Reforms that survive a change of government lower the cost of capital and enable financial close before market opportunities are competed away. Single-administration reforms struggle to mobilize long-duration capital.
- 2 Storage deployment can reshape wholesale markets in years, not decades.**
Once the regulatory window opens, capacity additions and price effects accumulate quickly. Day–night spreads compress, evening ramps soften, and the financiable universe expands within a single planning cycle.
- 3 Modularity — geographic and functional — requires regulation that matches it.**
BESS assets can relocate geographically and across market segments. Permitting regimes that accommodate this mobility (no full re-evaluation when adding storage to existing projects) and market classifications that allow cross-segment participation (revenue stacking between energy and capacity) unlock value that rigid frameworks foreclose.
- 4 Short-term distributional constraints — fiscal and tariff-related — are central to political feasibility.**
Market-based mechanisms avoid the political costs of direct subsidies and tariff stampings. But they require persistent and predictable price signals — which not every country can provide. Where distributional politics constrain market-based pricing, alternative pathways may be needed.
- 5 Wholesale-market success does not translate quickly to retail under PPA regimes.**
In Chile, regulated retail tariffs and long-term PPAs delay the passthrough of wholesale gains to end consumers. Countries with greater retail exposure to spot prices — Colombia, for example — may pursue more aggressive downstream strategies. Others may want to explore shortcuts.

Where to follow this work

PAPER

Full paper in preparation.

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